

Institutional Zone Mapper [JOAT]

// This Pine Script® code is subject to the terms of the Mozilla Public License 2.0 at <https://mozilla.org/MPL/2.0/>

// © officialjackofalltrades

//@version=6

indicator("Institutional Zone Mapper [JOAT]", "IZM [JOAT]", overlay=true,
max_boxes_count=500, max_labels_count=500)

// — Inputs

string G_OB = "Order Block Detection"

string G_ZONE = "Zone Settings"

string G_MIT = "Mitigation"

string G_VIS = "Visuals"

swingLen = input.int (7, "Swing Length", group=G_OB, minval=2, maxval=25)

obLook = input.int (20, "OB Candle Lookback", group=G_OB, minval=5, maxval=50)

dispMult = input.float(1.3, "Displacement ATR Mult", group=G_OB, minval=0.5,step=0.1)

maxOBs = input.int (5, "Max Active OBs / Side", group=G_OB, minval=1, maxval=20)

minStr = input.float(3.0, "Min Strength Filter", group=G_OB, minval=0.0,maxval=10, step=0.5)

zoneMult = input×float(0.75, "Zone ATR Width", group=G_ZONE, minval=0.1,step=0.05)

proxMult = input×float(0.25, "Proximity Buffer (ATR)", group=G_ZONE, minval=0.05,step=0.05,

tooltip="Highlight bar when close is within ATR*this of zone edge.")

posFilt = input.bool (false, "Position Filter", group=G_ZONE, tooltip="Show bull OBs only
below price, bear OBs only above price.")

mitAct = input.string("Stop Extending", "On Mitigation", group=G_MIT, options=["Keep","Stop
Extending","Delete"])

fadeMit = input.bool (true, "Fade On Mitigation", group=G_MIT)

showInfo = input.bool (true, "Show Info Labels", group=G_VIS)

showProx = input.bool (true, "Show Proximity Markers", group=G_VIS, tooltip="Diamonds when
price enters OB proximity zone.")

showDash = input.bool (true, "Show Dashboard", group=G_VIS)

bullCol = input×color(#60a5fa, "Bull OB Color", group=G_VIS)

bearCol = input×color(#fbbf24, "Bear OB Color", group=G_VIS)

// — Color helpers

```
dashBg = colorXnew(#0f172a, 8)
dashHd = colorXnew(#1e293b, 0)
tPrim  = colorXnew(#e2e8f0, 0)
tSec   = colorXnew(#64748b, 0)
```

```
// —— Custom type
```

```
type OB
  box   zoneBox
  label infoLbl
  float top
  float bot
  bool  isBull
  string session
  int   creationBar
  float strength
  bool  mitigated
  int   touchCount
```

```
// —— Arrays
```

```
var array<OB> bullOBs = array.new<OB>()
var array<OB> bearOBs = array.new<OB>()
```

```
// —— ATR
```

```
atrVal = taxatr(14)
```

```
// —— Session detection (UTC)
```

```
getSession(int t) =>
  h  = hour(t, "GMT")
  m  = minute(t, "GMT")
  mins = h × 60 + m
```

```
mins >= 480 and mins < 1020 ? "London" :  
mins >= 780 and mins < 1320 ? "NY" :  
mins >= 0 and mins < 540 ? "Asia" : "Off"
```

```
sessColor(string s) =>
```

```
s == "London" ? color.new(#60a5fa, 0) :  
s == "NY" ? color.new(#f472b6, 0) :  
s == "Asia" ? color.new(#34d399, 0) : color.new(#64748b, 0)
```

```
// ——— Strength scoring (0-10)
```

```
calcStrength(float disp, string sess, int age, float zh) =>
```

```
float ds = atrVal > 0 ? (disp / atrVal >= 3.0 ? 3.0 : disp / atrVal >= 2.0 ? 2.5 : disp / atrVal >= 1.5 ?  
2.0 : disp / atrVal >= 1.0 ? 1.5 : 1.0) : 1.0  
float ss = sess == "London" or sess == "NY" ? 2.0 : sess == "Asia" ? 1.0 : 0.5  
float as_ = age < 30 ? 2.0 : age < 75 ? 1.5 : age < 150 ? 1.0 : 0.5  
float zs = atrVal > 0 ? (zh / atrVal >= 0.5 and zh / atrVal <= 2.0 ? 3.0 : zh / atrVal >= 0.3 and zh /  
atrVal <= 3.0 ? 2.0 : 1.0) : 1.5  
math.min(10.0, ds + ss + as_ + zs)
```

```
// Strength → fill transparency (higher = more opaque)
```

```
strToAlpha(float s) =>
```

```
s >= 8.0 ? 78 : s >= 6.0 ? 83 : s >= 4.0 ? 88 : 93
```

```
// ——— Info label text
```

```
buildInfo(bool bull, float str_, string sess, int age, float zh, int tc) =>
```

```
pips = math.round(zh / syminfo.mintick)  
str.format("{0} ||| S:{1}/10 ||| {2} ||| Age:{3} ||| {4}p ||| T:{5}",  
bull ? "BULL OB" : "BEAR OB",  
str.toString(math.round(str_ * 10) / 10, "#.#"),  
sess, str.toString(age), str.toString(pips), str.toString(tc))
```

```
// ——— Lifecycle helpers
```

```

deleteOB(OB ob) =>
    box.delete(ob.zoneBox)
    label.delete(ob.infoLbl)

extendOBs(array<OB> arr) =>
    for ob in arr
        box.set_right(ob.zoneBox, bar_index + 30)
        if showInfo and not na(ob.infoLbl)
            label.set_x(ob.infoLbl, bar_index + 30)
            int age_ = bar_index - ob.creationBar
            label.set_text(ob.infoLbl, buildInfo(ob.isBull, ob.strength, ob.session, age_, ob.top - ob.bot,
ob.touchCount))

mitigateOBs(array<OB> arr) =>
    int i = 0
    while i < array.size(arr)
        ob = array×get(arr, i)
        bool mit = ob×isBull ? close < ob.bot : close > ob.top
        if mit and not ob.mitigated
            ob.mitigated := true
            if mitAct == "Delete"
                deleteOB(ob)
                array.remove(arr, i)
                i // explicit int return — matches type of other branches
            else if mitAct == "Stop Extending"
                box.set_right(ob.zoneBox, bar_index)
                box.set_extend(ob.zoneBox, extend.none)
                if fadeMit
                    col = ob×isBull ? bullCol : bearCol
                    box.set_bgcolor(ob.zoneBox, color.new(col, 96))
                    box.set_border_color(ob.zoneBox, color.new(col, 88))
                i += 1
            else
                if fadeMit
                    col = ob×isBull ? bullCol : bearCol
                    box.set_bgcolor(ob.zoneBox, color.new(col, 96))
                i += 1
    else
        else

```

```

// Update touch count
if close >= ob.bot and close <= ob.top
  ob.touchCount += 1
// Darken border on touch (visual feedback)
col_ = ob.isBull ? bullCol : bearCol
box.set_border_color(ob.zoneBox, color.new(col_, math.max(10, 60 - ob.touchCount *
8)))
i += 1

```

```

trimOBs(array<OB> arr) =>
  while array.size(arr) > maxOBs
    old = array.shift(arr)
    deleteOB(old)

```

```

// ——— Pivots

```

```

pivH = ta.pivohigh(high, swingLen, swingLen)
pivL = ta.pivotlow (low, swingLen, swingLen)

```

```

// ——— Bullish OB creation

```

```

if not na(pivL) and barstate.isconfirmed
  int  obOff = na
  float obDisp = na
  for j = 1 to obLook
    if not na(obOff)
      break
    if close[swingLen + j] < open[swingLen + j]
      obOff := swingLen + j
      obDisp := math.abs(close[swingLen] - close[swingLen + j])
  if not na(obOff) and not na(obDisp) and obDisp > atrVal * dispMult
    float zMid = math×avg(open[obOff], close[obOff])
    float zHalf = atrVal × zoneMult × 0.5
    float zTop = zMid + zHalf
    float zBot = zMid - zHalf
    bool passPF = not posFilt or zTop < close

```

```

string sess = getSession(time[obOff])
int  cBar = bar_index - obOff
float str_ = calcStrength(obDisp, sess, 0, zTop - zBot)
if str_ >= minStr and passPF
    col    = bullCol
    sCol    = sessColor(sess)
    alpha_ = strToAlpha(str_)
    b      = box.new(cBar, zTop, bar_index + 30, zBot,
        bgcolor=color.new(col, alpha_), border_color=sCol, border_width=2)
    lbl    = showInfo ? label.new(bar_index + 30, zTop, buildInfo(true, str_, sess, 0, zTop - zBot,
0),
        style=label.style_label_left, color=color.new(col, 100),
        textcolor=sCol, size=size_xsmall) :
        label.new(bar_index, zBot, "", color=color.new(color.white, 100), textcolor=color.white)
    array.push(bullOBs, OB.new(b, lbl, zTop, zBot, true, sess, cBar, str_, false, 0))
    trimOBs(bullOBs)

```

// ——— Bearish OB creation

```

if not na(pivH) and barstate.isconfirmed
    int  obOff = na
    float obDisp = na
    for j = 1 to obLook
        if not na(obOff)
            break
        if close[swingLen + j] > open[swingLen + j]
            obOff := swingLen + j
            obDisp := math.abs(close[swingLen] - close[swingLen + j])
    if not na(obOff) and not na(obDisp) and obDisp > atrVal * dispMult
        float zMid = math.avg(open[obOff], close[obOff])
        float zHalf = atrVal * zoneMult * 0.5
        float zTop = zMid + zHalf
        float zBot = zMid - zHalf
        bool passPF = not posFilt or zBot > close
        string sess = getSession(time[obOff])
        int  cBar = bar_index - obOff
        float str_ = calcStrength(obDisp, sess, 0, zTop - zBot)

```

```

if str_ >= minStr and passPF
  col    = bearCol
  sCol   = sessColor(sess)
  alpha_ = strToAlpha(str_)
  b      = box.new(cBar, zTop, bar_index + 30, zBot,
    bgcolor=color.new(col, alpha_), border_color=sCol, border_width=2)
  lbl    = showInfo ? label.new(bar_index + 30, zBot, buildInfo(false, str_, sess, 0, zTop - zBot,
0),
    style=label.style_label_left, color=color.new(col, 100),
    textcolor=sCol, size=size_xsmall) :
    label.new(bar_index, zTop, "", color=color.new(color.white, 100), textcolor=color.white)
  array.push(bearOBs, OB.new(b, lbl, zTop, zBot, false, sess, cBar, str_, false, 0))
  trimOBs(bearOBs)

```

// ——— Per-bar lifecycle

```

extendOBs(bullOBs)
extendOBs(bearOBs)
if barstate.isconfirmed
  mitigateOBs(bullOBs)
  mitigateOBs(bearOBs)

```

// ——— Touch detection + proximity

```

bool bTouch = false
bool sTouch = false
bool bProx  = false
bool sProx  = false
float proxDist = atrVal × proxMult

```

```

for ob in bullOBs
  if close >= ob.bot and close <= ob.top
    bTouch := true
  else if close > ob.top and close <= ob.top + proxDist
    bProx := true

```

```

for ob in bearOBs

```

```

if close >= ob.bot and close <= ob.top
  sTouch := true
else if close < ob.bot and close >= ob.bot - proxDist
  sProx := true

// Proximity markers
plotchar(showProx and bProx ? close : na, "Bull Prox", "◆",
  location.absolute, color.new(bullCol, 15), size=sizextiny)
plotchar(showProx and sProx ? close : na, "Bear Prox", "◆",
  location.absolute, color.new(bearCol, 15), size=sizextiny)

// ——— Dashboard

```

```

if showDash
  var table d = table.new(position.top_right, 2, 9,
    bgcolor=dashBg, border_color=color.new(#1e293b, 0), border_width=1,
    frame_color=color.new(#334155, 0), frame_width=1)
  string sessNow = getSession(timenow)
  table.cell(d, 0, 0, "IZM [JOAT]", text_color=tPrim, text_size=size×small, bgcolor=dashHd)
  table.cell(d, 1, 0, syminfo.ticker + " " + timeframe.period, text_color=tSec, text_size=size×small,
    bgcolor=dashHd)
  table.cell(d, 0, 1, "Bull OBs", text_color=tSec, text_size=size×small,
    bgcolor=color.new(#0f172a, 0))
  table.cell(d, 1, 1, str.tostring(array.size(bullOBs)), text_color=bullCol, text_size=size×small,
    bgcolor=color.new(#0f172a, 0))
  table.cell(d, 0, 2, "Bear OBs", text_color=tSec, text_size=size×small,
    bgcolor=color.new(#0f172a, 0))
  table.cell(d, 1, 2, str.tostring(array.size(bearOBs)), text_color=bearCol, text_size=size×small,
    bgcolor=color.new(#0f172a, 0))
  table.cell(d, 0, 3, "Bull Touch", text_color=tSec, text_size=size×small,
    bgcolor=color.new(#0f172a, 0))
  table.cell(d, 1, 3, bTouch ? "ACTIVE" : bProx ? "NEAR" : "—",
    text_color=bTouch ? bullCol : bProx ? color.new(bullCol, 35) : tSec, text_size=size×small,
    bgcolor=color.new(#0f172a, 0))
  table.cell(d, 0, 4, "Bear Touch", text_color=tSec, text_size=size×small,
    bgcolor=color.new(#0f172a, 0))
  table.cell(d, 1, 4, sTouch ? "ACTIVE" : sProx ? "NEAR" : "—",

```



```

text_color=sTouch ? bearCol : sProx ? color.new(bearCol, 35) : tSec, text_size=size×small,
bgcolor=color.new(#0f172a, 0))
table.cell(d, 0, 5, "Session", text_color=tSec, text_size=size×small,
bgcolor=color.new(#0f172a, 0))
table.cell(d, 1, 5, sessNow, text_color=sessColor(sessNow), text_size=size×small,
bgcolor=color.new(#0f172a, 0))
table.cell(d, 0, 6, "ATR", text_color=tSec, text_size=size×small, bgcolor=color.new(#0f172a,
0))
table.cell(d, 1, 6, str.tostring(atrVal, format.mintick), text_color=tPrim, text_size=size×small,
bgcolor=color.new(#0f172a, 0))
table.cell(d, 0, 7, "Disp Min", text_color=tSec, text_size=size×small,
bgcolor=color.new(#0f172a, 0))
table.cell(d, 1, 7, str.tostring(atrVal * dispMult, format.mintick), text_color=tPrim,
text_size=size×small, bgcolor=color.new(#0f172a, 0))
table.cell(d, 0, 8, "Min Score", text_color=tSec, text_size=size×small,
bgcolor=color.new(#0f172a, 0))
table.cell(d, 1, 8, str.tostring(minStr, "#.#") + "/10", text_color=tPrim, text_size=size×small,
bgcolor=color.new(#0f172a, 0))

```

// ——— Alerts

```

alertcondition(bTouch and barstate.isconfirmed, "IZM: Price at Bull OB", "Price retesting a
Bullish Order Block")
alertcondition(sTouch and barstate.isconfirmed, "IZM: Price at Bear OB", "Price retesting a
Bearish Order Block")
alertcondition(bProx and barstate.isconfirmed, "IZM: Approaching Bull OB", "Price
approaching a Bullish Order Block")
alertcondition(sProx and barstate.isconfirmed, "IZM: Approaching Bear OB", "Price
approaching a Bearish Order Block")
alertcondition(not na(pivL) and barstate.isconfirmed, "IZM: Bull OB Candidate", "New Bullish
Order Block zone created")
alertcondition(not na(pivH) and barstate.isconfirmed, "IZM: Bear OB Candidate", "New Bearish
Order Block zone created")

```